Brian Takita

LinkedIn: linkedin.com/in/briantakita | **Email:** btak.dev@gmail.com | **GitHub:** btakita
Greater Boston, MA | **Website:** briantakita.me | **Phone:** +1 (424) 249-2350

SUMMARY

Builds **agentic** developer infrastructure where **humans and agents work in parallel**. Most recently at a **Confidential AI Startup**, did **full-stack** development of real-time AI systems and **closed-loop agentic workflows** with **Next.js**, **LiveKit**, **Python**, and **Rust**. In open source, created **agent-doc** — a **document-as-interface** for persistent AI sessions with an **agent-agnostic core** and **pluggable LLM backends** — and other open-source libraries in **Rust**, **Python**, **Zig**, and **TypeScript**. Earlier architected a production **real-time** multimodal AI avatar pipeline at **Presence AI** on **NVIDIA H100 GPUs** via **WebRTC**. A decade of **context engineering** across **reactive** and **event-driven** systems.

SKILLS

AWS, **AWS** CDK, **AWS** Lambda, Agentic Systems, **Agents**, Angular, Azure, C#, **CI/CD**, Claude, Context Engineering, Convex, CopilotKit, Django, **Docker**, DuckDB, FalkorDB, FastAPI, Firebase, Flask, Flutter, Go, Google Cloud, Grafana, GraphQL, Graphene, Heroku, Jira, **MongoDB**, Next.js, **Node.js**, NumPy, Pandas, **PostgreSQL**, PyTorch, **Python**, REST, Ray, React, Ruby on Rails, **Rust**, SQLAlchemy, SQLite, Spec-Driven Development, Supabase, SvelteJS, Terraform, TimescaleDB, **TypeScript**, Vue.js, WebRTC, **ZeroMQ**, Zig, Zod, msgspec

CAREER HIGHLIGHTS

- **Presence AI** — Architected real-time multimodal AI avatar pipeline — text, audio, and video lip-sync on NVIDIA H100 GPUs via WebRTC
- **Open Source** — corky: git-native correspondence in Rust | agent-doc: document-as-interface for AI agents | RSpec: BDD core contributor | rr: pioneered proxy-based test doubles (now `patch` in Python)
- **MonsterRodHolders** — Rescued a failing WordPress/WooCommerce storefront and grew MRR 50% in one month via SEO and Google Ads
- **Pivotal Labs** — Mentored developers across 20+ startups. Speaker at RailsConf 2008. Early contributor to Behavior Driven Development
- **TrueCar / honk.com** — Led vehicle review platform development at honk.com (acquired by TrueCar); senior product team at TrueCar
- **Milyoni** — Social Entertainment Platform for Warner Brothers, Lions Gate, Paramount Studios, Universal Music Group
- **Censible** — 8-year ESG analytics platform — Martin Investments analyst portal, ESG Portfolio Rebalancing system
- **Context Engineering** — Domain ontology system (extensible) since 2014 — structuring agent reasoning across domains
- **Reactive Frameworks** — 15-year lineage: backbone-signal → ctx-core → rmemo → relysjs — applied to modern agentic systems

EXPERIENCE

INDEPENDENT CONSULTING — *SOFTWARE ENGINEER / CONSULTANT*

2026-02 – Present (2026 – Present) | Remote

- Confidential AI Startup — Real-time AI systems engineering (Apr–May 2026): built real-time voice AI and generative media pipelines — low-latency streaming inference and agent-harness analysis tooling — and delivered investor-facing demos under tight timelines. Rust, Python, TypeScript, LiveKit, WebRTC, CopilotKit.
- Equity Funding Source — Architected and built a full-stack loan origination platform on Cloudflare's edge stack: public prequalification intake, D1-backed application-lead pipeline, and multi-role authentication (admin/borrower/broker/investor) via Cloudflare Access with one-time PIN — no external auth vendor. Built an invite-only admin portal for lead review and account management, and designed an OCR document-intelligence pipeline with reviewed field extraction feeding loan-servicing integration. Cloudflare Workers, D1, R2, Cloudflare Access, Hono, React, TypeScript, Resend, Wrangler.
- MonsterRodHolders.com — Rescued a failing WooCommerce storefront (57-SKU catalog) and grew MRR 50% in one month through SEO and Google Ads. Diagnosed a card-testing bot attack (170+ fraudulent PayPal orders in two weeks, threatening merchant-account standing) from order-database analysis and mitigated it — deployed Cloudflare Turnstile on checkout to block headless bots, removed the low-cost product the bots used to validate stolen cards, and drove PayPal fraud controls (3D Secure, AVS/velocity filters). Hardened WordPress security after an admin-account compromise, configured Cloudflare CDN for performance and DDoS protection, and restored analytics tracking. WordPress, WooCommerce, Cloudflare Turnstile, PayPal, SEO, Google Ads, Cloudflare CDN, Google Analytics.

PRESENCE AI — *LEAD BACKEND AI ENGINEER*

2024-04 – 2026-02 (1 yr 10 mos) | Remote

- Built real-time AI avatar pipeline processing video/audio streams with sub-100ms latency
- Designed reactive signal graph architecture for deterministic avatar response timing
- Implemented WebRTC streaming infrastructure for live avatar sessions
- Developed multimodal AI pipeline integrating LLM, TTS, lip-sync models, image generation, and live video/audio/text avatars
- Created context engineering system for agentic workflow orchestration
- Implemented ZeroMQ-based event-driven messaging architecture for real-time avatar pipeline coordination
- Scaled the avatar pipeline with Ray for distributed and real-time parallel compute — orchestrating PyTorch lip-sync inference and NumPy numeric workloads, with msgspec for low-latency message serialization

OPEN SOURCE — *CREATOR / SOLO DEVELOPER*

2006 – Present | Remote

- Built corky: git-native correspondence system in Rust syncing email, Telegram, and Slack into scoped markdown mailboxes with per-contact agent context
- Built and deployed Google Cloud Platform applications including a public correspondence kit with OAuth2 + Gmail API integration and private data management tools
- Created agent-doc: document-as-interface for persistent AI conversations with CRDT-based streaming merge for real-time human+agent co-editing
- Built agent-kit: Rust SDK for CLI tools integrating with AI agent loops — session management, tmux pane routing, instruction file discovery
- Created tmux-router: declarative tmux pane routing in Rust for multiplexed agent orchestration across concurrent development contexts
- Built tagpath: cross-language identifier analysis tool in Rust with convention detection, semantic search across 14 tree-sitter grammars

- Built tsift: hybrid code-search engine in Rust (BM25 + vector search) over tree-sitter-parsed codebases, with a provider-neutral graph-store layer — SQLite reference backend plus experimental DuckDB/DuckPGQ and Ladybug backends evaluated through a parity-and-performance promotion gate
- Created lazily-rs/lazily-py/lazily-zig: reactive signals with cross-language ports maintaining consistent semantics across Rust, Python, and Zig
- Created ctx-core: explicit, scalable cross-platform reactive contexts in JavaScript — foundation for rmemo, RelementJS, rappstack
- Created rr (double Ruby): pioneered proxy-based test doubles (mock.proxy, stub.proxy) — pattern now standard as patch in Python's pytest/unittest
- Core contributor to RSpec: the Ruby Behavior Driven Development testing framework
- Created domain ontology system (existence-lang, formerly philosophy, 2014): structuring agent reasoning across domains with extensible scope, context, and resolution — gives agents and humans a consistent vocabulary across code, correspondence, and data contexts

BRIAN TAKITA — *FULL-STACK ENGINEER*

2019-01 – 2026-02 (7 yrs 1 mo) | Remote

- Menu.com: Multi-tenant restaurant ordering and payment platform — consumer web ordering, restaurant back-office, and Cordova mobile app. Stripe tokenization for payment processing. Developed AWS CDK + Lambda serverless functions integrated with AWS Fargate Windows .NET API on ECS
- BestInform: European information platform — pixel-perfect Figma implementation, real-time support messaging. Angular 14, PrimeNG, Firebase
- Guard App: Security service platform — admin portal and mobile apps for patrol management, incident reporting, NFC verification. Flutter, Angular, Firebase, Java Spring Boot
- Drivemond: Taxi and ride-hailing platform for Chicago — passenger and driver mobile apps with real-time matching and GPS tracking. Flutter, AWS
- Galler AI: AI-powered image generation and editing platform — React frontend with Figma design system; cropping, clipping, drawing tools
- SafeSchoolz (Aegis Platform): AI safety platform detecting firearms and alerting authorities — built gun detection model (~85% mAP). Android, .NET, Python/Jupyter, Twilio
- IIPM: Membership and certification platform for Irish pension specialists — member and admin portals. WordPress, PHP, JavaScript/jQuery, Stripe

MILYONI, INC. — *ARCHITECT / LEAD DEVELOPER*

2012-03 – 2013-03 (1 yr 1 mo) | Pleasanton, CA

- Lead developer for Social Entertainment Platform including video, social marketing, gamification, real-time social interaction, and interactive engagements synced with video
- Platform used by Warner Brothers, Lions Gate, Paramount Studios, Universal Music Group, Hearst Media, Starz, Ovation TV, Astro, Focus Features, Funimation, Sundance Now
- Mentored junior & mid-level developers; proponent of flattening communication across the organization
- Implemented event-driven architecture and AWS IAM-based access control for platform services

TRUECAR / HONK.COM — *LEAD DEVELOPER → SENIOR SOFTWARE ENGINEER*

2009-08 – 2012-02 (2 yrs 7 mos) | Santa Monica, CA

- Led development of honk.com, a vehicle review and white-label platform used by USAA and Wall Street Journal Autos — acquired by TrueCar
- Partnered on product vision for social car shopping that was key to the acquisition
- Senior member of advanced product team at TrueCar — introduced new products, partner integrations (WSJ.com, USAA.com), and devops automation
- Chef scripts for server and workstation automation; high availability deployment on EC2 and Engine Yard

- Implemented MongoDB-based Car Buyer social media experience logging system for tracking user engagement across the platform

CENSIBLE — *LEAD PRODUCT DEVELOPER*

2016-01 – 2024-04 (8 yrs 4 mos) | Greater New York City Area

- Built Martin Investments analyst portal with ESG rating algorithms, XLSX export handling 10,000+ rows, and SVG/PDF report generation
- Built ESG Portfolio Rebalancing system and ESG analytics platform
- Processed ESG and portfolio datasets with Python and Pandas for the Martin Investments analyst portal — rating computation, data normalization, and report-ready aggregations across large holdings datasets

PIVOTAL LABS — *AGILE ENGINEER*

2006-01 – 2009-08 (3 yrs 7 mos)

- Mentored developers across 20+ startups, bootstrapping their engineering process, team, and initial product
- Speaker at RailsConf 2008
- Leadership role in common code, practices, and open source (Erector, JS Test Server, Unison)
- Agile Development in the XP tradition: TDD, Pair Programming, Daily Standups, Retrospectives
- Implemented AWS IAM access control and functional reactive programming patterns across client projects

EDUCATION

University of the Pacific BS, Engineering Physics (1997 – 2002)

CERTIFICATIONS

- Advanced Permaculture Course in Teaching
- Permaculture Design Certificate

LANGUAGES

English: Native